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INHALED INTERNAL HEAT/THERMAL CONDUCTION THERAPY FOR ESOPHAGEAL CANCER AND RELATED ESOPHAGEAL PRE-CANCEROUS DISEASES

Abstract

A method for treating esophageal cancer in situ with heated air and by extension heat that can be inhaled through a face mask, at an elevated temperature less than the highest sauna air temperatures and at a constant positive air pressure. The inhaled heated air will heat the very thin tracheoesophageal party wall membrane between the esophagus and the respiratory trachea in the throat. The tracheoesophageal party wall membrane therefore can heat the esophageal cancer tumor adjacent to the heated trachea and tracheoesophageal party wall membrane. The heated air is inspired/inhaled through the mouth and into the trachea/windpipe. Then the trachea and trachea tube under heated constant positive pressure will then transfer heat through and to the esophagus wall, and esophagus and tumor. The heat from the heated air inspiration through the trachea migrates/transmits/disperses across the membrane separating the trachea/windpipe from the esophagus and hits the esophagus with heat.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation-in-part of application Ser. No. 19/041,841 filed Jan. 30, 2025, which '841 application is a continuation of application Ser. No. 18/801,599, filed Aug. 12, 2024, now U.S. Pat. No. 12,233,213 B1 dated Feb. 25, 2025, which '599 application is a continuation-in-part of application Ser. No. 18/600,704, filed Mar. 9, 2024, now U.S. Pat. No. 12,083,040 issued Sep. 10, 2024, which '704 application is a continuation-in-part of application Ser. No. 18/432,843, filed Feb. 5, 2024, now U.S. Pat. No. 12,004,993 B1 issued Jun. 11, 2024, for a **TOPICALLY APPLIED HEAT/THERAPY FOR SKIN LESIONS AND OTHER DISEASES**, which '843 application is a continuation-in-part of application Ser. No. 18/418,179, filed Jan. 19, 2024, which '599, '704, '843 and '179 applications are incorporated by reference herein. Applicant claims priority under 35 U.S.C. § 120 from the '599, '704, '843 and '179 applications.

INCORPORATION BY REFERENCE

[0002] This application incorporates by reference in its entirety Applicant's U.S. Pat. No. 12,233,213 issued Feb. 25, 2025, entitled **INHALED INTERNAL HEAT/THERMAL CONDUCTION THERAPY FOR ESOPHAGEAL CANCER**.

[0003] This application also incorporates by reference in its entirety Applicant's U.S. Pat. No. 12,004,993 B1 issued Jun. 11, 2024, entitled **TOPICALLY APPLIED HEAT/THERAPY FOR SKIN LESIONS AND OTHER DISEASES**, and which '993 Patent was filed under USPTO application Ser. No. 18/432,843 on Feb. 5, 2024. This application also incorporates by reference in its entirety Applicant's U.S. Pat. No. 11,234,861 B2 issued Feb. 1, 2022, entitled **RESPIRATORY THERAPEUTIC ELECTRIC HEAT SOURCE FACE MASK**, and which '861 Patent was filed under USPTO application Ser. No. 17/337,352 on Jun. 2, 2021, which '352 application was published as US 2021/0282964 A1 on Sep. 16, 2021. This application also incorporates by reference in its entirety Applicant's U.S. Pat. No. 10,905,585 B1 issued Feb. 2, 2021, entitled **RESPIRATORY THERAPEUTIC ELECTRIC HEAT SOURCE FACE MASK**, and which '585 Patent was filed under USPTO application Ser. No. 16/893,212 on Jun. 4, 2020.

FIELD OF THE INVENTION

[0004] The present invention relates to treating in-situ esophageal cancer, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, with heated air that can be inhaled through a heat source, such as, for example, a heat gun inhaler device, at an elevated

temperature less than the highest sauna air temperatures. The inhaled heated air will heat the very thin tracheoesophageal party wall, which is synonymous with the membranous trachea membrane located between the esophagus and the respiratory trachea windpipe, extending in and below the throat. The tracheoesophageal party wall membrane therefore can heat the esophageal cancer tumor adjacent to the heated trachea and tracheoesophageal party wall membrane, without the need for sedation of the patient during the invasive procedures currently used surgically.

BACKGROUND OF THE INVENTION

[0005] All references are included in their entirety as if reproduced in full herein.

[0006] In Cleveland Clinic, Esophageal Cancer, 2024, esophageal cancer is defined as tumors within the esophagus between the mouth and stomach of a person and being either adenocarcinomas in the lower part of the esophagus and Squamous cell carcinomas that affect the squamous cells lining the esophagus. Symptoms and causes of esophageal cancer are discussed, as are diagnostic tools for detecting esophageal cancer. Treatment regimens are discussed, including surgery, radiation therapy, chemotherapy, endoscopic submucosal dissection (ESD) or endoscopic mucosal resection (EMR), endoscopic laser therapy, photodynamic therapy (PDT), targeted HER2 protein therapy and immunotherapy. Lifestyle alternatives are also discussed to reduce the possibility of esophageal cancer.

[0007] Christodoulidis et al, in “Advancements and challenges in the treatment of esophageal cancer: A comprehensive review,” World Journal of Clinical Oncology, Dec. 24, 2024 15(12): 463-1467, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11514372/> states that “Esophageal cancer (EC) is an aggressive malignancy with a poor prognosis, ranking seventh in incidence and sixth cancer-related deaths globally.” It also notes that early-stage EC is difficult to diagnose, leading to late-stage diagnoses with poor outcomes. About 20% of patients live 5 yrs. 5 yr. survival ranges from 5% to 47%. Therefore, there is a dismal, huge unmet medical need for new efficacious, non-toxic, non-invasive cost-effective treatments.

[0008] Mount Sinai Esophageal Cancer, 2024, discusses esophageal cancer in general.

[0009] American Cancer Society (ACS) in “Treating Esophagus Cancer,” 2024, discloses local and systemic treatment options for patients with Esophageal Cancer.

[0010] Mayo Clinic, in Esophageal cancer, 2024, discusses various treatment regimens for treating esophageal cancer.

[0011] Esophageal cancer can be treated with heat in situ. See National Cancer Institute “Hyperthermia to treat cancer,” at cancer.gov/about-cancer/treatment/types/hyperthermia. NCI discloses that temperature at about 113° F. kills cancer cells with little or no harm to normal human body tissue. The New York Times, opus cited, at page 3/9, discloses that treating human tissue topically at 50° C., for 30 seconds 122° F., kills the parasite/Leishmaniasis protozoa deep inside the human tissue.

[0012] See also Daniel Yetman, “Hyperthermia Treatment for Cancer: Uses and Effectiveness,” Healthline Newsletter, Nov. 9, 2022 (reviewed by Teresa Hagan Thomas). Yetman discloses that temperatures of 106° F. to 111° F. are toxic to cancer cells while sparing normal cells.

[0013] Saunas have been used for thousands of years. See Wikipedia at “The Finnish Sauna”. The earliest versions date from 7000 BC. Millions of people have used them in the intervening years with great safety. The sauna room, which is the sauna, is generally warmed to 176-230° F., op cit. The sauna room exposes the entire body to the heat.

[0014] “The Physics Factbook”, edited by Glen Elert, discloses different ranges of temperatures of Sauna units, at about 176-212° F., 150-194° F., 170-180° F., and 160° F.

[0015] “Emma, in “Ideal Sauna Temperatures: How Hot Is Your Sauna?,” website, 2020 discloses the Finnish sauna temperature is typically 160-194° F. for 30 to 45 minutes. It also discloses that “the American standard implies temperatures from 160 to 194° F., but in many European countries, allowed temperatures are in the range of 160 to 220° F.” and, “However, the Finnish Sauna Society recommends that the temperature in the sauna should be from 176 to 194° F. (80-90° C.) with the

ultimate maximum of 212° F. (100° C.”).

[0016]] text missing or illegible when filed

[0017] text missing or illegible when filed Thermoregulation” Healthline, discloses that sweat cools human skin as it evaporates, and that the hypothalamus sends signals to various organs and systems in the body.

[0018] The skin's immense blood supply helps regulate temperature: i.e., dilated vessels/capillaries near the surface of the skin allow for heat loss through the blood, through sweating, while constricted vessels retain heat. The skin regulates body temperature with its blood supply, elevated by temperatures, which are expelled through sweat from the skin with its large surface area. Humidity affects thermoregulation by limiting sweat evaporation and thus heat loss.

[0019] Without being limited, held, or bound, to any particular theory or mechanism of action, heat through hot air and positive pressure is the inventive step, this is the critical inventive step of the invention. For example, the sauna will not raise the temperature of lung cancer tissue very much, if at all, as it is analogous to pouring water into a bucket with no bottom. A typical heated sauna room lacks positive pressure, therefore with respiration, gaseous exchange occurs between the inspired air adding oxygen, and the exhaled air containing 4-6% carbon dioxide, which is a greenhouse gas trapping heat, and exhaling trapped heat and carbon dioxide occurs through normal respiration exhalation. In the sauna, the entire skin is exposed to heat, opening the pores to expel sweat and heat derived from the enormous blood supply trafficking throughout the body and skin. Because of the lack of positive pressure, a patient can remain in the sauna until the cows come home without raising the temperature much or at all of tissues and tumor cells.

[0020] However, in contrast, the Applicant's invention isolates the trachea and esophagus, separated by a membranous trachea, so that inhaled heated air provided with positive air flow pressure, will traverse the party wall into the adjacent esophagus. The inventor has tried the invention upon himself for up to one hour at 180° F. to 190° F., without toxicity.

[0021] Likewise, the heated air can alternatively heat the respiratory system from the rest of the body and skin so that the respiratory and heated lung tissue generated through the face mask, along with positive air flow pressure, will disperse/exit more slowly, allowing an elevated tissue temperature in the lung. Instead of having a bucket with no bottom, now there is a bucket with holes in the bottom to drain water out more slowly, allowing a higher ambient air into the respiratory system, where the carbon dioxide generated through normal respiration trapping heat is no match for the abundant forced positive heat and hot air of the invention.

[0022] Moreover, “National Cancer Institute” in Hyperthermia in Cancer Treatment” discloses: “Hyperthermia (also called thermal therapy or thermotherapy) is a type of cancer treatment in which body tissue is exposed to high temperatures (up to at least 113° F.). Research has shown that high temperatures can damage and kill cancer cells, usually with minimal injury to normal tissues (1). By killing cancer cells and damaging proteins and structures within cells (2), hyperthermia may shrink tumors.”

[0023] This is the “holy grail” of cancer medicine that is most earnestly pursued or sought after. Despite many decades of research, exploitable differences between normal cells and cancer cells remain elusive. Very few, if any, differences exist in tumor and normal cell antigens for targeting. The massive difference in the uptake of glucose by tumor cells against normal cells is one. This resulted in the well-known Positive Emission Tomography (PET Scan). Drugs are currently being designed to support this major exploitable difference in glucose uptake. The differential in sensitivity to heat is another holy grail of cancer medicine, which the Applicant utilizes to treat tumors and dysplasia in the lung.

[0024] Cleveland Clinic, Oct. 18, 2018, discloses “Hyperthermia: Why Heat Can Make Cancer Treatments More Potent”. They use the application of heat at 109-110° F. with radiation and chemotherapy.

[0025] Moreover, Katsarelias, et al, in “The Effect of Temperature and perfusion time on Response,

Toxicity and Survival in Patients with In-transit Melanoma Metastases Treated with Isolated Limb Perfusion”, Ann Surg Oncol: 2018; 25(7): 1836-1842 May 15, 2018, discloses a well-established technique to administer “a very high dose” of chemotherapy, at elevated temperature, to isolated tumor sites without causing overwhelming systemic damage.

[0026] Also, the Cleveland Clinic, in “Hyperthermia: Why Heat Can Make Cancer Treatment More Potent—Combining heat with chemo or radiation can shrink tumors” Cleveland Clinic Cancer Care website, Oct. 18, 2018, states that heat can be combined with chemotherapy or radiation therapy to reduce or destroy cancer tumor cells.

[0027] Additionally, NIH National Cancer Institute, in “Hyperthermia in Cancer Treatment,” August 31, 2011, disclosed that hyperthermia cats in combination with chemotherapy and/or radiation can damage cancer cells and enhance the anti-tumor effects of chemotherapy and/or radiation.” Further disclosed is that at 113° F., tumor cells are harmed and killed, leaving normal cells unscathed.

[0028] Applicant's prior art U.S. Pat. No. 11,234,861 B2 issued Feb. 1, 2022, entitled RESPIRATORY THERAPEUTIC ELECTRIC HEAT SOURCE FACE MASK discloses administering controlled heat from a heat gun through a mask, to provide heated air to treat respiratory diseases.

[0029] Raiko et al, in Imaging based internal body temperature measurements, in the Journal of “temp-toolbox” in TEMPERATURE, Vol 7, no. 4 at KTMP 71769006, discloses non-invasive temperature measurements of internal body tissues.

OBJECTS OF THE INVENTION

[0030] It is therefore an object of the present invention to provide heated air and by extension heat that can be inhaled through a heat source, such as a heat gun inhaler device, at an elevated temperature less than the highest sauna air temperatures for treatment of esophageal cancer, abnormal esophageal pre-cancerous cells and Barrett's esophagus syndrome with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus. The inhaled heated air will heat the very thin tracheoesophageal party wall membrane between the esophagus and the respiratory trachea in the throat. The tracheoesophageal party wall membrane therefore can heat the esophageal cancer tumor adjacent to the heated trachea and tracheoesophageal party wall membrane, without the need for sedation of the patient during the invasive procedures currently used surgically.

[0031] The heat from the heater air inspiration through the trachea migrates/transmits/disperses across the membrane separating the Trachea/windpipe from the esophagus and hits the esophagus with heat.

[0032] It is also very desirable to have the heated air being delivered by a handheld, battery powered or powered by 120 V utility power heat gun with safety locks, operable for heat treatment of esophageal cancers in real time, which is way less toxic, less painful, more efficacious, quicker, and which operates at a higher temperature, and is cost effective. Moreover, the higher temperatures available facilitate treatment of esophageal cancerous tumors by delivering heated air through a face mask directly delivering the heated air to a cancer patient's throat tracheas so that the heated air will conductively heat the cancerous esophageal tumor by passing through the tracheoesophageal party wall membrane, to the esophageal tumor in the esophagus, on the adjacent opposite side of the tracheoesophageal party wall membrane.

[0033] It is also an object to provide a cost-effective battery or AC powered heated air delivery machine, and where the cost of use is a small fraction of the cost of existing invasive endoscopic devices, and without costly maintenance and with less invasive existing surgery, radiation, and/or chemotherapy.

[0034] Other objects of the invention will become apparent from the following description of the present invention.

SUMMARY OF THE INVENTION

[0035] Esophageal cancer is the 7th ranked cancer in incidence globally and is the 6th leading cause of death from cancer.

[0036] The present invention for the treatment of esophageal cancer, includes a mechanism of action that includes a novel, non-invasive use of hyperthermia, which in general has been FDA approved since 1987. Tumor cells are proven sensitive and susceptible to heat killing, by exposure to heat of at least 113 degrees F., optionally up to the highest level of sauna air, such as 180+degrees F.

[0037] For example, the Second Law of Thermodynamics states that heat flows downhill from hotter to colder. In the present invention, tolerable inhaled heated air flows from the trachea windpipe of the patient to the adjacent membranous trachea party wall separating the trachea from the esophagus, and thereby, by heat conduction. Heat conduction is the transfer of heat between objects that are touching, on one side is the heated air in the trachea under positive pressure about 180° F., on the other side is the esophagus at about 98.6° F., both sides touching the thin membranous trachea transferring heat into the esophagus, killing any cancerous tumors, or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues in the esophagus adjacent to the adjacent membranous trachea party wall.

[0038] Conductivity also, causes heat from the inspired hot air in the trachea to be conducted into the esophagus through the membranous tracheoesophageal wall.

[0039] With respect to the dimensions of the esophagus subject to heat conductivity, the esophagus is about 10-13 inches long. The diameter of the esophagus is about 1 inch, and the circumference is about 3.14 inches from the mathematical calculation of $\text{Pi} \times \text{diameter}$ of an object. The membranous trachea is $\frac{1}{3}$ the circumference which is about 1 inch.

[0040] The membranous trachea is therefor about 3 to 8 mm, or, 0.1181 to 0.314961 inches, about $\frac{1}{8}$.sup.th inch to about $\frac{1}{4}$ inch in thickness, which is perfectly thin to transfer heat from the trachea into the esophagus, through the membranous trachea/tracheoesophageal party wall. Since the tracheoesophageal party wall parallels the esophagus the conductive, cytotoxic heat extends through the entire length of the esophagus.

[0041] Heat from the inspired hot air in the trachea is provided under positive pressure through the membranous trachea located between the trachea and the esophagus, which heats up the membranous trachea and transmits conductive heat into the esophagus, in the vicinity of a cancerous esophageal tumor.

[0042] The conductive heat extends through the whole length of the esophagus. The inhaled conductive heat kills the one or more tumors afflicting the esophagus, because heat provided under positive pressure from the heat source hits the entire length of the esophagus.

[0043] It is estimated that heat will be applied in a predetermined time duration, as prescribed by the patient's physician, such as up to ABOUT one hour OR more or less per treatment. The heat conductivity treatment is repeated if necessary.

[0044] Therefore, the present invention relates to treating in situ esophageal cancer, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, with heated air that can be inhaled through a heat source, such as a heat gun inhaler device, at an elevated temperature less than the highest sauna air temperatures. The inhaled heated air will heat the very thin tracheoesophageal party wall membrane between the esophagus and the respiratory trachea in the throat. The tracheoesophageal party wall membrane (also known as the "membranous trachea") therefore can heat by conduction the esophageal cancer tumor, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, adjacent to the heated trachea and tracheoesophageal party wall membrane, without the need for sedation of the patient during the invasive procedures currently used surgically.

[0045] The heat from the heater air inspiration through the trachea migrates/transmits/disperses heat via conduction across the tracheoesophageal party wall membrane separating the

trachea/windpipe from the esophagus and hits the esophagus with heat.

[0046] The heated air will be inspired/inhaled through the mouth and into the trachea/windpipe. Then the trachea and trachea tube under heated constant positive pressure will then transfer heat through and to the esophagus wall, and to the esophagus, itself with one or more esophageal tumor, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus.

[0047] The esophagus parallels the trachea/windpipe, the wall of the trachea is very close to the wall the inner wall of the esophagus. Both share the tracheoesophageal party wall, also known as the membranous trachea

[0048] Then it follows by inhaling/inspiring heated air up to 180°+degrees F. with the invention, it is possible enough heat will emanate/traffic by heat conduction from the trachea/windpipe wall and conductively interface with the esophagus wall and esophagus. Therefore, the inhaled heat within the trachea migrates across the tracheoesophageal party wall membrane, separating the trachea/windpipe from the esophagus and via conduction the inhaled heat in the trachea/windpipe raises the heat of the localized esophageal tumor at a threshold of about 113° F., which will kill cancer in the esophagus, as well as abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues.

[0049] As disclosed in Raiko_Imaging-based-internal-body-temp-measurements-the-journal-temp-toolbox_TEMPERATURE_Vol7no4_KTMP_7_1769006, in order to non-invasively and remotely measure the temperature of the cancer cells being killed and annihilated, caused by pre-determined time exposure to the conducted heat through the membrane separating the trachea/windpipe from the esophageal tumor site within the esophagus, external probes using external image-based internal body temperature measurements can be administered, such non-invasive devices as ultrasound, magnetic resonance imaging, computed tomography, microwave radiometry, photoacoustic imaging and near-infrared spectrometer, either within a hollow cavity such as CT scan or MRI imaging tunnel machine, or by a hand-held device such as an ultrasound transducer used in Applicant's U.S. Pat. No. 11,234,861 B2 "Respiratory Therapeutic Electric Heat Source Face Mask", dated Feb. 1, 2022, and its parent U.S. Pat. No. 10,905,585 B1 "Respiratory Therapeutic Electric Heat Source Face Mask", dated Feb. 2, 2021, are both incorporated by reference herein in their entirety, as if reproduced in full for information about heated air sources, such as heat guns, with air temperature safety locks and interlocks providing controlled and monitored application of hot air into the respiratory system of a patient, without exceeding a predetermined safe temperature threshold of hot air. These heat guns disclosed in these patents can also be used, although they require a 120 Volt plug in power supply.

[0050] To treat esophageal tumors adjacent to the tracheoesophageal party wall membrane separating the trachea from the esophagus, the Applicant's device can be placed directly over the patient's mouth to introduce treated air to the trachea. The method of treatment includes the steps of; [0051] a) placing a concentrated but efficacious quantity of heated air generated by a heat source through a face mask having an outlet tube within or at the mouth, or before the trachea of a patient, adjacent to the tracheoesophageal party wall membrane, spaced from an esophageal cancer or pre-cancerous esophageal disease, on and/or extending within the esophagus of the patient; [0052] b) directing the efficacious applied heated air from the heat source into the trachea, to conductively transfer the heated air of the heat source, through the tracheoesophageal party wall membrane, thereby directly to the esophageal cancer tumor or pre-cancerous esophageal disease, adjacent thereto; [0053] c) applying the efficacious exterior topically applied heated air directed from the heat source to a full area of the cancer or pre-cancerous esophageal disease on or within the esophagus; [0054] d) programming the heat source for producing and directing the efficacious exterior topically applied heated air having a temperature sufficient for, and for a sufficient time, at a constant air pressure, to destroy the cancer or pre-cancerous esophageal disease on and/or extending within the esophagus of the patient; [0055] e) monitoring the heat source and constant air

pressure for insuring that the efficacious inhaled heated air being delivered to the cancer or pre-cancerous esophageal disease through the tracheoesophageal party wall membrane does not exceed a predetermined safe temperature threshold of the inhaled heated air; and, [0056] f) mounting a display on or adjacent the heat source showing the set programmed temperature and a current temperature of the heated air delivered to the esophageal tumor cancer or pre-cancerous esophageal disease being treated, the cancer or pre-cancerous esophageal disease being located within or on the esophagus in the vicinity of the tracheoesophageal party wall membrane separating the esophagus from the trachea of the patient.

[0057] Therefore, the heat from the heated air traverses through the trachea and conductively migrates/transmits/disperses across the tracheoesophageal party wall membrane separating the trachea/windpipe from the esophagus and directly contacts the esophagus with heat at a constant air pressure, which kills esophageal cancer, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus.

[0058] The method further includes the heat source preferably being a heat gun with an outlet nozzle sufficiently large and shaped to deliver the heated air through the face mask into the patient's trachea, wherein a display on the heat gun comprises an LCD screen showing the temperatures, including the set programmed temperature of the heated air. The set programmed temperature is displayed on an upper portion of the heat gun, which preferably has one sized nozzle, preferably one sided nozzle to connect to the silicone tubing to the mouth of the patient from the heat source for delivery of the applied heated air to the cancer located on or within the esophagus of the patient, at or adjacent to the surface of the tracheoesophageal party wall membrane of the patient, preferably in the range of about 113° F., to about 180°+ F., where 113° F. has been described in the literature known to those skilled in the art of cancer treatment as being the minimal temperature for subjecting the cancerous esophageal tumor to cell death.

[0059] A further option includes the step of inserting a subcutaneous needle probe temperature monitor into the in-situ site of the esophageal tumor being treated and measuring the temperature of the tumor being treated.

[0060] Because coffee and hot chocolate are served between 160° F.-185° F. of heat to billions of people (i.e., at Starbucks or other hot beverage retail establishments) for many decades without burning the mouth, then it is reasonable that Applicant's invention can be applied at 140°+ F. of heat through the trachea of the patient to the tumor or cancer.

[0061] Applicant has heard a number of times whereby a surgeon opens up a cancer patient's torso, and gasps nothing to be done, extensive disease, or looked at an x ray with the same conclusion. Now, with the applicant's invention, the surgeon can avoid surgically opening a patient's chest, abdomen, etc., and using the applicant's invention, he or she can kill the esophageal cancer cells, in a brief time, since the application of heated air at 113° F. is known to kill tumor cells, as well as abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus.

[0062] As taught by Raiko, et al, op cit., the method of Applicant's invention includes an option of using a temperature monitored application of hot air to the transfer of the patient, where the threshold temperature of the patient's esophageal tumor is measured and monitored by heat gun temperature sensor features of the heated air, when exiting the heat gun during the procedure, in order to inactivate the tumor by tumor cell death or to inactivate pre-cancerous esophageal disease, such as dysphagia and Barrett's disease.

[0063] An example of a preferred embodiment for the heat source is a heat gun, which uses nozzles for aiming monitored hot air flow, with built-in safety locks and interlocks, but which also has a built-in laser infrared temperature sensor and LCD display, such as is manufactured by Master Heat Tools, as MASTER PROHEAT STC®, which is a surface temperature control heat gun. This heat gun can dispense hot air through the face mask tube to the trachea of a patient at specified exiting

air temperatures, so that the esophageal cancer or pre-cancerous esophageal disease are treated at a predetermined temperature to inactivate the cancerous tumor or pre-cancerous esophageal disease, which are located adjacent to the tracheoesophageal party wall membrane separating the trachea through which the hot air is delivered, from the esophagus, which is conductively heated by the heated air in the trachea transversing therethrough to the adjacent esophageal tumor or pre-cancerous esophageal disease.

[0064] The preferred range of the esophageal tumor temperature being treated can be preferably in the range of about 113° F. up to about 180+° F., such as, for example, about 120° F. to about 150° F.

[0065] Therefore, the present invention preferably utilizes no invasive endoscopic device, with possible pain and internal injuries. Second, the heated air derived from the Applicant's inventions covers the entire esophageal tumor at one time, beginning to end, without invasively needing sedation and an endoscopic probe within the patient's esophagus.

[0066] In contrast to endoscopic procedures, the present invention monitors and regulates the temperature of the esophageal tumor being treated. The heated air flow of the Applicant's invention may be raised and lowered by pressing control buttons on the heat source, which is preferably a heat gun with built in safety locks to prevent inadvertent excess heat and air pressure being administered to the esophageal tumor adjacent to the aforementioned tracheoesophageal party wall membrane of the patient.

[0067] For example, Applicant's heat gun can be, preferably, battery powered and portable. If utility power such as 120 Volts is available, the heat gun can be powered by utility power as well.

[0068] In summary, the aforementioned administration of applicant's invention herein includes the inhaled administration of temperature controlled heated air to the trachea of the patient, with a concentrated inhaled heat source, for destroying esophageal tumors located in the esophagus adjacent to the tracheoesophageal party wall membrane separating the trachea of the patient from the closely adjacent esophagus of the patient, as well as abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus.

[0069] As noted above, the inhaled hot air applied to the trachea of the patient is adjusted upward to a temperature of the esophageal tumor being treated, where the temperature is adjusted upward from about 120° F. to about 130° F., to have a net temperature of about at least 113° F. at the esophageal tumor.

[0070] In a preferred embodiment for treating esophageal cancer, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, the method steps further including the following details: [0071] a) install a correct nozzle on the heat gun; [0072] b) determine and adjust the temperature of the airflow; [0073] c) program and provide the heated hot air flow at a threshold temperature of an esophageal tumor sufficient to remove the esophageal tumor, abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus; [0074] d) adjust the output nozzle of applicant's heat gun and supply hot air through the patient's mouth and trachea to a location of the thin tracheoesophageal party wall membrane separating the trachea from the esophagus in the vicinity of the esophageal tumor and abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus; [0075] e) activate the temperature-controlled heat gun with locks and interlocks, at which time the entire tumor will be conductively heated across the thin tracheoesophageal party wall membrane from the trachea into the esophagus with heat derived from hot air at constant air pressure and controlled threshold temperatures; and, [0076] f) monitoring the temperature of the heated air for ensuring that the heated air does not exceed a safe temperature threshold within the trachea and esophagus of the patient.

[0077] As noted in Applicant's prior art U.S. Pat. No. 11,234,861 B2, while the power source for

the heat gun is preferably AC power, optionally the power source can alternatively be a low voltage DC power source, such as a battery, or the power source can be AC power transformed to a low voltage DC power source.

[0078] Preferably, the prior art of Sabin '861 heat gun heats air to a sufficiently high temperature is between about at least 80 degrees F. and about 275 degrees F., optionally, in which the sufficiently high temperature is at least 113 degrees F. up to about 230 degrees F. Also, optionally, the humidity is preferably about 90 to 95%, which is maintained by human respiration in the throat of a person being treated.

[0079] In order to keep the pressure of the heated air at a tolerable level compatible with human respiration, the air pressure of the heated air is compatible with typical air pressure flows in a Continuous Positive Air Pressure (CPAP) machine. Most CPAP machines pump air in the range from 6 to 15 cm/H₂O (centimeters of water pressure), such as, for example, an air flow is set at 8 cm/H₂O.

[0080] To convert that to air pressure measured in millimeters of mercury (Hg), one divides the cm/H₂O amount by a factor of 1.36. Therefore, a CPAP air flow pressure of 8 cm/H₂O is divided by a factor of 1.36, wherein $8/1.36=5.88$ Hg. Stated in psi (pounds per square inch), an 8 cm/H₂O=5.88 Hg, or 2.88 psi.

[0081] Therefore, the positive pressure applied to the heated air flow is between about 3 psi to about 16 psi, or expressed in centimeters per unit of water, as being from 1-20 cm/H₂O.

[0082] Even at the upper limit of 15, the air pressure being pumped from a CPAP machine into the person is only 11 Hg, or 5.4 psi.

[0083] Since typical heat guns are not designed to expel the heated air at such human tolerable air pressures that are regularly set in CPAP machines, there is a needed determination to make sure that the heated air, which is heated at a predetermined temperature comparable to cause cell death of the esophageal tumor, for example, or at sauna levels of 150° F. or above, is pumped into through a heat resistant-type tube (rated for high temperatures) and into the person's improved face mask disclosed in this application, at air pressure of no more than 11 Hg, or 5.4 psi at the upper level of human respiratory tolerance, or preferably a lower amount, such as for example, at 5.88 Hg or 2.88 psi.

[0084] For safe inhalation at human respiratory rates, the volume of air must be controlled down from the high rates capable of being expelled by a heat gun in industrial applications, down to an acceptable level tolerable by human respiration. For example, tidal volume (symbol V_{sub.T} or TV) is the typical human lung volume, representing the normal volume of air displaced between normal inhalation and exhalation when extra effort is not applied. In a healthy, young human adult, the tidal volume is approximately 500 mL per inspiration or 7 mL/kg of body mass. Furthermore, a normal minute volume, while resting, is about 5-8 liters per minute in humans. Minute volume generally decreases when at rest and increases with exercise. For example, during light activities, minute volume may be around 12 liters. Since heat guns are capable of producing heated air at about 190 liters per minute, the air volume can be titrated down to 5-8 liters per minute, or 12 liters per minute, or 50 liters per minute, whichever is best for the person being treated.

[0085] In the present invention, the prior art of Sabin '861 heat gun must have the capability of providing heated air in the range of 80° F. to 275° F., preferably at sauna heat temperature levels of about 113 degrees F. to about 180+degrees F., and at constant air pressure levels of no more than about 5.4 psi for human respiratory tolerance. While any heat gun which is capable of the aforementioned temperature and air pressure range limitations, non-limiting examples of such heat guns include the Master Pro Heat Gun models 1400 and 1500.

[0086] The Master heat gun model PH-1400, which operates at 120/230 V AC, outgoing air temperatures can be as low as 113 degrees F., extending up to 1000 degrees F. The Master heat gun model has a finger-operable ON/OFF switch on a trigger position of the handle, which has three settings depending upon selected movement of the switch. In a "HEAT" position, the top of the

switch is depressed inward. To achieve a “COOL” position, the switch is set so that the top and bottom portions are equally extended. To turn the switch off, the bottom of the switch is completely pushed inward.

[0087] The Master Pro Heat gun model PH-1400 can also be set at a human tolerable upper limit, such as for example 200 F, if medically appropriate, at air flow volumes of as little as 4 CFM (cubic feet per minute). The rear of the heat gun handle has a pair of turnable knobs, where the left knob is twisted and turned to adjust the airflow up or down to predetermined human tolerable levels, which are displayed in an LCD screen above or adjacent to the knobs. To adjust the temperature of the emitted air, the right knob is twisted and turned to raise or lower the designated heat of the air flow output. A set temperature is displayed for several seconds, until the actual output air temperature is displayed on the LCD screen.

[0088] A further safety prior art of Sabin '861 feature in the Master Pro Heat gun model PH-1400 is that adjacent to the two knobs there is provided a locking lever or key, typically but not necessarily magnetic, that is pushed or moved to set and lock the output air temperature and pressure. The settings inputted by the two knobs cannot be changed if the LCD indicates a “Lock” icon, due to the manipulation of the locking lever or key. The locking lever or key can alternatively be used for converting the temperature from Fahrenheit units to Celsius.

[0089] Alternatively, the prior art of Sabin '861 Master heat gun model PH-1500, also operates at 120/230 V AC, outgoing air temperatures can be as low as 113 degrees F., extending up to 1000 degrees F. The Master heat gun model 1500 also has a finger-operable ON/OFF switch on a trigger position of the handle, which has three settings depending upon selected movement of the switch. In a “HEAT” position, the top of the switch is depressed inward. To achieve a “COOL” position, the switch is set so that the top and bottom portions are equally extended. To turn the switch off, the bottom of the switch is completely pushed inward.

[0090] For use in provided heated air through a heat resistant flexible hose to a disclosed herein face mask covering the nose and mouth of the person, the prior art of Sabin '861 Master Pro Heat gun model PH-1500 can also be set at a human tolerable upper limit, such as for example 200 F or even 300 F, if medically appropriate, at air flow volumes of as little as 4 CFM (cubic feet per minute). The rear of the heat gun handle also has an LCD screen, but adjacent to, or below, the LCD screen there is provided an array of programmable settings with keypad up/down arrow keys, where the left up and down arrow keys are manipulated to adjust air pressure and the right up and down arrow keys are manipulated to adjust the temperature up or down to predetermined human tolerable levels, which are displayed in an LCD screen above or adjacent to the key pad. In the middle between the left and right-side arrow keys are two other keys, where the upper key has a “P” inscribed thereon, for program the desired inputted air pressure and temperature outputs. The lower middle key is used to change the output levels from Fahrenheit to Celsius, or vice versa.

[0091] A further safety feature in the prior art of Sabin '861 Master Pro Heat gun model PH-1500 is the programmable keys can be used to set and lock the output air temperature and pressure. The settings inputted by the two knobs cannot be changed if locked by the programmable keys.

[0092] In a further alternate embodiment, in a hospital setting, the prior art of Sabin '861 heat gun can be installed on a mobile I.V. stand, with more than one heat gun attached to an individual I.V. stand, or collapsible folding cart, with an AC wall outlet or, optionally, to a Honda generator on the cart for full ambulatory use (when in ventilated conditions to eliminate carbon monoxide), which lasts over eight hours at $\frac{1}{4}$ power. It can last over three hours at full power, and by dividing the pressurized heated air, can treat two people with one machine. For example, the conversion of CFM to liters per minute is 28.32; thus, a heat gun at 120° F. heat at 3.5 CFM times the factor of 28.32=99.12 liters per minute. Since a human takes 5-8 liters at rest, therefore the present invention can bleed off 99.12 minus 8=91.12 liters per minute, for treating in two persons simultaneously.

[0093] As described in the prior art of Sabin '861, other optional safety controls include air pressure controllers, to keep the heated air at those lower levels of pumped air pressure, so that the incoming

heated air flow is also limited to “human tolerable air pressure”, which typically may be no more than about at 15 cm/H₂O, or 11 Hg, or 5.4 psi, or within the CPAP acceptable range of air pressure of 6 to 15 cm/H₂O, or 4.4 Hg to 11 Hg, or 2.16 psi to 5.4 psi.

[0094] As described in the prior art of Sabin '861, for those low human tolerable air pressure flows, optionally there may be provided a microprocessor-controlled air pressure distributor associated with the heat gun, so that the psi of the outgoing air pressure can be maintained at those low human tolerable air pressures, regardless of the heat of the air.

[0095] In general, since the device is a medical device, optimally the prior art of Sabin '861 heat guns have air temperature and air pressure controls, such as thermostats for temperature control, with automatic shutoff features and interlock. For example, the Master Pro Heat gun has a minimal CFM of 4 and a safety interlock built-in for temperature and air flow stopping. Converted to liters=0.28 Liters per minute, this is below the air flow of a human at rest with normal activities, which is about 5-8liters per minute.

[0096] In connection therewith, while the prior art of Sabin '861 heat guns may have finger operable keys for controlling air temperature and air pressure, optionally these person operable controls and shutoffs may also have a simple AC wall outlet plug-in adapter that acts as a fail-safe automatic shutoff and interlock preventing any heat output above a threshold maximum and preventing any air pressure above a threshold maximum, such as tolerated in a CPAP machine, to prevent accidental increases of temperatures and air pressures above what is tolerable in human respiration.

[0097] Since a heat gun is a powerful device plugged into potentially lethal voltage and can deliver powerful airflow and temperatures to 1000° F., Applicant has tested temperatures at the tip of the heated air output, at a distance of one inch out therefrom, of the heat gun when on full heat, in replicate six+times. When the heat gun claims to emit heated air at 1200° F., Applicant measured the actual heated air from the heat gun to be about 487° F. Therefore, the listed manufacturing of heat guns' base temperature on the red-hot nozzle tip at a distance of about two inches from the tip of the heat gun is widely exaggerated. However, it may be desirable in certain environments to use a safety interlock AC outlet plug-in adapter, to separately monitor the temperature and pressure delivered to a patient and to shut down the system in case either the pressure or temperature in the patient hose exceed predetermined safety limits of heated air and pressure, tolerable in human respiration of air into the trachea adjacent to the membranous party wall separating the trachea from the esophagus, and then, by heat conduction, into the wall of the esophagus afflicted with one or more esophageal cancerous tumors.

[0098] As also described in the prior art of Sabin '861, the aforementioned simple interlock plug-in adapter is a fail-safe shutoff to completely shut the heat gun down if the heat gun itself has a malfunction in its operation or use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0099] The present invention can best be understood in connection with the accompanying drawings, it is noted that the invention is not limited to the precise embodiments shown in the following drawings, in which:

[0100] FIG. 1 is a perspective view of a healthcare provider preparing to administer heated air to a patient.

[0101] FIG. 2 is a close-up detailed view of the LCD viewing screen, taken along dashed view circle line “2” of FIG. 1, showing the temperature of the exiting air from the heat gun through a nozzle.

[0102] FIG. 3 is a close-up detailed view of an alternate embodiment for a heat gun with a dual

LCD viewing screen, displaying both the target temperature selected on the top of the LCD screen, and below, the current moving threshold temperature of the exiting heated air on the bottom of the LCD screen.

[0103] FIG. 4 is a front perspective view of the heat gun of FIG. 1, provided in a carry case, with a set of nozzles of varying exiting diameters.

[0104] FIG. 5 is a front-perspective view of the prior art of Sabin '861 inhalation delivery of heated air for treatment of respiratory illnesses, as in FIG. 1.

[0105] FIG. 6 is a side-perspective view of the prior art of Sabin '861 inhalation delivery of heated air for treatment of respiratory illnesses, as in FIG. 1.

[0106] FIG. 7 is a cross-sectional cutaway view of the prior art of Sabin '861 inhalation mask for delivery of heated air for treatment of respiratory illnesses, showing heated air flow through the mouth of a patient.

[0107] FIG. 8 is a medical scan image showing in dashed lines the thin (which is about $\frac{1}{8}$ inch to $\frac{1}{4}$ inch) tracheoesophageal party wall membrane separating the trachea from the adjacent esophagus in a medical patient so that inhaled heat can be transferred via conduction across the tracheoesophageal party wall membrane from the trachea into the esophagus, wherein the prior art of Sabin '861 masks of FIGS. 5-7 herein are used to deliver heated air into the trachea of the medical patient with esophageal cancer. The heat from the heater air inspiration through the trachea migrates/transmits/disperses across the membrane separating the Trachea/windpipe from the esophagus and hits the esophagus with heat. FIG. 8 also shows the robust, sturdy, cartilaginous tissue in the posterior of the view, in dashed lines.

[0108] FIG. 9 is a diagrammatic side view in partial cross section of the throat, trachea (windpipe), and esophagus, including the thin membranous trachea-esophageal party wall that separates the trachea from the esophagus, wherein, by heat conduction, inhaled heated air in the trachea passes through the membranous trachea-esophageal party wall by heat conduction to the esophagus which may have one or more esophageal cancerous tumors or pre-cancerous cells therein.

DETAILED DESCRIPTION OF THE INVENTION

[0109] The drawings are for illustrative purposes only, and the preferred mode for carrying out the invention is described herein.

[0110] As used throughout this specification, the word “may” is used in a permissive sense (i.e., meaning having the potential to, or being optional), rather than a mandatory sense (i.e., meaning must), as more than one embodiment of the invention may be disclosed herein. Similarly, the words “include,” “including,” and “includes” mean including but not limited to.

[0111] The phrases “at least one,” “one or more,” and “and/or” may be open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “one or more of A, B, and C”, and “A, B, and/or C” herein means all of the following possible combinations: A alone; or B alone; or C alone; or A and B together; or A and C together; or B and C together; or A, B and C together.

[0112] Also, the disclosures of all patents, published patent applications, and non-patent literature cited within this document are incorporated herein in their entirety by reference. However, it is noted that the citing of any reference within this disclosure, i.e., any patents, published patent applications, and non-patent literature, is not an admission regarding a determination as to its availability as prior art with respect to the herein disclosed and claimed apparatus/method.

Furthermore, any reference made throughout this specification to “one embodiment” or “an embodiment” means that a particular feature or characteristic described in connection therewith is included in at least that one particular embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Therefore, the described features, advantages, and characteristics of any particular aspect of an embodiment disclosed herein may be combined in any suitable manner with any of the other embodiments disclosed herein.

[0113] In connection with the present invention, the drawings show one or more embodiments, but the present invention is not limited to that shown in the drawings.

[0114] FIG. 1 also shows a typical heat gun HG, which can have temperature control locks and interlocks SL (such as explicitly disclosed in Applicant's '861 patent and '585 patent, incorporated by reference herein, in its entirety), to keep the outflowing hot air HAEX to cool down to the therapeutic temperature TLe of the skin being treated of about 105° F.-180°.sup.+ F., optionally, in a range of 105° F.-135° F., preferably at about 120° F. to about 130° F. for a time period up to about one hour, depending on physician determination of duration. While the LCDA of the heat gun HG displays the programmed temperature of the heated air flow HAF required to cause heat conduction through the party wall separating the trachea windpipe from the esophagus, to achieve the actual required temperature in an esophageal tumor.

[0115] FIG. 2 shows the LCD viewing screen LCDA of heat gun HG, showing the temperature TAEX of the exiting air flow HAF, from the heat gun HG through a nozzle N. The LCD viewing screen LCDA displays the temperature of the heat HAEX exiting from the heat gun HG.

[0116] FIG. 3 shows the preferred embodiment for a heat gun HG-STC with a dual LCD viewing screen LCD-STC, displaying the programmed selected surface temperature on the top of the LCD-STC screen, (i.e. 120 degrees F.), in order to achieve the desired threshold temperature TLe (i.e. 120 degrees F.) of the internal esophageal tumor treated, displayed on the bottom of LCD screen LCD-STC.

[0117] With respect to use and operation of the Master ProAir STC (identified herein as HG-STC), with a built-in contactless laser infrared temperature thermometer, Applicant incorporates by reference in its entirety, the Instruction Manual thereof, submitted in the Information Disclosure Statement (IDS) filed herein, which discusses the simultaneous use of the heat gun with its built-in contactless temperature thermometer for measuring surface temperatures of exiting heated air measuring exited temperature to treat esophageal tumor. The Instructional Manual also discusses choice of nozzles, using a trigger switch and trigger lock, how to operate the gun pressing the trigger switch for the heating element and fan startup, aiming the laser infrared light beams at the target to be measured upon heat application thereto, engaging and releasing the trigger lock, using the PROLOC supervisory key to change between supplying heated air and measuring exited temperature and using the surface temperature controls, along with explanatory diagrams associated therewith.

[0118] When administering inhaled heat through the trachea of a patient, the temperature of the esophageal tumor being treated by heat conduction through the thin tracheoesophageal party wall membrane, can be measured via hand held devices, such as ultrasound transducers, in a similar manner to the hand held temperature monitors for measuring skin surface temperature, where the temperature data can be displayed in real time during administration of the inhaled heated air, such as disclosed in Raiko et al, in Imaging based internal body temperature measurements, in the Journal of "temp-toolbox" in TEMPERATURE, Vol 7, no. 4 at KTMP 71769006, discloses non-invasive temperature measurements of internal body tissues.

[0119] FIG. 4 shows the heat gun HG or HGSTC of FIG. 1, provided in a carry case kit K, with a set of nozzles N of varying exiting diameters. FIG. 4 also shows a plurality of masking templates 12, 13, 14, 15, each having respective holes 12a of varying area sizes, fitting to different sized skin lesions not used with the present invention.

[0120] FIG. 5 shows an alternate embodiment for the prior art of Sabin '861 inhalation delivery of heated air for treatment of respiratory illnesses.

[0121] FIG. 6 also shows the prior art of Sabin '861 inhalation delivery of heated air for treatment of respiratory illnesses, as in FIG. 1.

[0122] FIG. 7 shows the prior art of Sabin '861 inhalation mask for delivery of heated air for treatment of respiratory illnesses, showing heated air flow through the mouth of a patient. FIG. 7 is a detail cross sectional view showing the attachment of the prior art of Sabin '861 semi-rigid

silicone tube extending through the foam barrier, which has a distal heated air discharge end and a proximal end joinable via a joint to the insulated flexible conduit, which is connected to the heat source. The patient's lips interface with the foam barrier at the distal end of the rigid silicone tube. The opening in the foam barrier of the prior art of Sabin '861 device shown in FIG. 7 matches the opening of the orifice of the ½ inch to about ¾ inch outer diameter (OD) sized silicone tube, while the lips interface with the foam, so that the heated air traffics from the hollow inside ½ inch internal diameter (id) orifice of the tubing, flowing directly to the open orifice of the user's mouth, so that the heated air is totally bypassing direct contact with the lips, nose and face of the user.

[0123] Not shown in prior art of Sabin '861 FIG. 7 is a detachable manometer or to measure air pressure, which quick connects to the opening where the tubing exits the foam.

[0124] Preferably, the face mask heats air to a sufficiently high temperature is between about at least 80° F. and about 275° F., optionally, which the sufficiently high temperature is at least 113 degrees F. up to 180+degrees F. at a relative humidity of from about 90% to about 95% relative humidity.

[0125] In order to keep the pressure of the heated air at a tolerable level compatible with human respiration, the air pressure of the heated air is compatible with typical air pressure flows analogous to a Continuous Positive Air Pressure (CPAP) machine. Most CPAP machines pump air in the range from 6 to 15 cm/H2O (centimeters of water pressure), such as, for example, an air flow is set at 8 cm/H2O.

[0126] However, unlike CPAP machines with provide air to both the nose and throat, the mask of the present invention is limited to provide heated air only to the mouth and throat of the patient, so that the heated air provided by gentle positive air pressure to the trachea, will transfer by heat conduction though the membranous party wall separating the trachea and the esophagus having one or more esophageal cancerous tumors therein.

[0127] In the present invention, the prior art of Sabin '861 heat gun HG shown in FIG. 1, must have the capability of providing heated air in the range of 113 d ° F. to 180+° F., preferably at sauna heat temperature levels of about 150 to 200° F. up to about 230° F., and at air pressure levels of no more than about 5.4 psi for human respiratory tolerance. While any heat gun which is capable of the aforementioned temperature and air pressure range limitations, non-limiting examples of such heat guns include the Master Pro Heat Gun models 1400 and 1500.

[0128] In the preferred alternate embodiment shown in in prior art of Sabin '861, at FIGS. 5-7, the face mask **800** includes a resilient, compressible barrier **810**, made of polyurethane foam or other resilient compressible material, to provide a barrier between the mouth of the user inhaling hot air from heat.

[0129] While heat at approximately 130° F. can be tolerated by the lips of the user's mouth, but when the heat is raised to about 160° F. or above, the lips, face and/or eyes of the user can be injured, parched, or scorched from direct contact with the heated air.

[0130] However, when the prior art of Sabin '861 barrier **810** is provided in the form of a cylinder of foam of between 2 and 4 inches depth, preferably about 3 inches in depth, that depth provided isolates the lips from direct contact with the heated air, but permits the heated air to flow directly into the open mouth of the user, and directly into the respiratory system, without reducing efficacy.

[0131] To hold the prior art of Sabin '861 mask **810** with the compressible, resilient barrier **810**, the barrier **810** is held in place in a collar **812** attached to head gear **814**, which wraps around the skull of the user. The heated air from the heat source **838** trends through flexible conduit tubing **840**, which is wrapped within an insulated sleeve **842**. A temperature gauge **850** includes a sensor **853** piercing the flexible conduit **840** near the connection between the piercing the flexible conduit **840** near the connection between the silicone rubber tubing **820** within the barrier **810**, where a cable **852** transmits the temperature detected by the sensor **853** to the thermometer **850** attachable to the insulating sleeve **842** of the heated air conduit **840**. The thermometer **850** is also a Wi-Fi

transmitter to a receiver, which has a safety beeping and flashing if set temperature is exceeded. An example of a thermometer is ThermoPro TP-07 cooking thermometer. Semi rigid straight silicon tubing **820** is preferable 2-4 inches in length with an outer diameter of between $\frac{3}{4}$ inch and one inch, with $\frac{1}{2}$ inch internal diameter (id) is ideal diameter. A quick release optional CPAP air pressure gauge can be provided to attach to the opening of the tube or/and a detachable manometer can also be used. The straight silicon tubing **820** is provided with joint shoulder collar **824**, to facilitate connection to a flexible conduit tubing **840** from the heat source **838**.

[0132] Therefore, the prior art of Sabin '861 heat mask **800** with the foam barrier **810** is a new, heat-resistant soft barrier mouthpiece, as depicted in FIGS. 5-7, with a $\frac{1}{2}$ inch internal diameter (id) heat emitting orifice of the tubing matching the open mouth of the patient, bypassing the lips, face, eyes, nose, cheeks, etc.

[0133] The prior art of Sabin '861 soft barrier mouthpiece mask **800**, includes a cylindrical, face-conforming barrier block **810** of polyurethane foam (or optionally viscoelastic foam), which has more or less a $\frac{5}{8}$ inch through-hole, in which the silicone tube **820** is attached proximally to the flexible hole **811**, within block **810**, and attached to conduit **840**, connected to the heat gun heat source.

[0134] Moreover, the depth of the prior art of Sabin '861 foam cylindrical tubing **820** is 2 to 4 inches, preferably about 3 inches and the barrier cylinder block **810** has a diameter of about 5 inches, more or less.

[0135] Because the hot air comes out of the flexible tube **840** from heat source **838**, and then goes through the straight silicone tube **820** therewithin, the user's lips are spaced apart from the exiting hot air of 113-180+° F., coming from the flexible conduit **840** attached to the heat source **838**, and the heated air is advanced through the straight silicone tubing **820** within the foam barrier cylindrical sleeve **810**, directly into the respiratory airway and lungs of the user, while bypassing the FACE, lips of the user, because the lips surround the exterior circumference of the distal end of the straight silicone tubing **820**, and the lips are not directly exposed to the hot air passing through the straight silicone tube **820**, as the hot air is directed into the mouth orifice of the respiratory system.

[0136] A wearable strap assembly **812**, **814** is provided with the cylindrical foam barrier **810** of the mask/mouthpiece **800**, so that it stays comfortably on the face and skull of the wearer.

[0137] The prior art of Sabin '861 straight silicone tubing **820** is, for example, a Tygon 3355-1 Silicone tubing $\frac{1}{2}$ ' inner diameter (id) and $\frac{3}{4}$ ' outer diameter (OD), with a $\frac{1}{8}$ ' thick wall, heat resistant to 400° F., or other suitable tubing.

[0138] As in prior art FIGS. 5, 6 and 7, of Sabin '861, the distal nozzle of heat gun **838** mates in a joint **832**, including a nipple **834**, wherein the nozzle of heat gun **838** is insertable within the nipple **834**, which engages the open proximal end **836** of flexible tubing **840**. At its opposite distal end, the flexible tubing **840** is connected to, and insertable within, the inner diameter of the proximal end of straight silicon tubing **820**, extending within foam barrier **810** of mask **800**, shown in FIGS. 8 and 9.

[0139] The heat source **838** contains internal and/or inline safety interlocks to monitor the temperature and the pressure of the heated and pressurized air delivered to a patient and to shut down the heat source **838** if the air temperature or the pressure of the air to be delivered to the patient exceeds preselected safety limits. For example, if one or more the predetermined safety limits is exceeded, a switch operates, so that the heat gun will not operate and a visual display with display indicates an "OFF" mode. The interlock may also include a mechanism for locking in a preferred predetermined temperature and pressure range capable of inactivating the virus causing COVID-19 disease (SARS-CoV-2).

[0140] Optional safety goggles (not shown) may be provided to further protect the eyes of the user. A remote wireless monitor **831** with safety alarms is installed to further monitor temperature and as a safety feature.

[0141] FIG. 8 is a medical scan image showing in dashed lines the thin tracheoesophageal party wall membrane separating the trachea from the adjacent esophagus in a medical patient so that inhaled heat can be transferred via conduction across the thin tracheoesophageal party wall membrane from the trachea into the esophagus, wherein the prior art of Sabin '861 masks of FIGS. 7-10 are used to deliver heated air into the trachea of the medical patient with esophageal cancer. FIG. 8 also shows the robust, sturdy, cartilaginous tissue in the posterior of the view, in dashed lines.

[0142] The heat from the heater air inspiration through the trachea migrates/transmits/disperses across the tracheoesophageal party wall membrane separating the patient's trachea/windpipe from the esophagus and hits the esophagus with heat.

[0143] The heated air will be inspired/inhaled through the mouth and into the trachea/windpipe. Then the trachea and trachea tube under heated constant positive air pressure will then conductively transfer heat through and to the esophagus wall, and to the esophageal tumor.

[0144] This is because the esophagus parallels the trachea/windpipe, and the wall of the trachea is very close to the wall the inner wall of the esophagus.

[0145] Then it follows by inhaling/inspiring heated air of at least about 113° F. with the invention, it is possible enough heat will emanate/traffic from the trachea/windpipe wall and interface with the esophagus wall and esophagus. Therefore, the inhaled heat provided at constant air pressure within the trachea migrates across the tracheoesophageal party wall membrane separating the trachea/windpipe from the esophagus and via conduction the inhaled heated air in the trachea/windpipe raises the heat of the localized esophageal tumor at a threshold of about 113° F., which will kill cancer cells and abnormal esophageal pre-cancerous cells associated with Barrett's esophagus, with dysplasia cells and tissues in the esophagus.

[0146] In order to non-invasively measure the temperature of the cancer cells being annihilated by cell death caused by pre-determined time exposure to the conducted heat through the membrane separating the trachea/windpipe from the esophageal tumor site within the esophagus, optionally external probes using external image-based internal body temperature measurements described in Raiko et al, op cit., can be administered such non-invasive devices as ultrasound, magnetic resonance imaging, computed tomography, microwave radiometry, photoacoustic imaging and near-infrared spectrometer, either within a hollow cavity such as CT scan or MRI imaging tunnel machine, or by a hand-held device such as an ultrasound transducer used in combination with a pulsed laser light source provided built into the heat source heat gun, or via a separate handheld measurer such as an ultrasound transducer, where the raised temperature within the tissues of the esophageal tumor can be measured and displayed on a display device, such as on the heat gun or on a display of the remote handheld device or imaging tunnel device.

[0147] The prior art of Sabin '861 masks of FIGS. 5-7 herein are used for treating esophageal cancer in situ with heated air that can be inhaled at an elevated temperature less than the highest sauna air temperatures. The inhaled heated air will heat the very thin tracheoesophageal party wall membrane between the esophagus and the respiratory trachea in the throat.

[0148] The thin tracheoesophageal party wall membrane therefore can heat the esophageal cancer tumor or abnormal esophageal pre-cancerous cells associated with Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, adjacent to the heated trachea and tracheoesophageal party wall membrane, without the need for sedation of the patient during the invasive procedures currently used surgically. The heated air will be inspired/inhaled through the mouth and into the trachea/windpipe. Then the trachea and trachea tube under heated constant positive pressure will then transfer heat through and to the esophagus wall, and thence to the esophagus to inactivate the esophageal tumor or other abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues.

[0149] The esophagus parallels the trachea/windpipe, the wall of the trachea is very close to the

wall the inner wall of the esophagus. Then it follows by inhaling/inspiring about 113° F. with the invention, it is possible enough heat will emanate/traffic from the trachea/windpipe wall, through the thin tracheoesophageal party wall membrane and interface with the esophagus wall and esophagus. The inhaled heat, when provided at a temperature of about 113° F. will kill cancer in the esophagus.

[0150] Although the temperature of 113° F. will kill the cancer in the esophagus, the heat can be provided at a higher amount, up to about 180+° F. to accommodate any heat loss caused by the presence of the thin tracheoesophageal party wall membrane separating the trachea from the esophagus. However, because of the thinness of the tracheoesophageal party wall membrane separating the esophagus from the trachea, it is assumed that a sufficient amount of the heat will be transferred through the tracheoesophageal party wall membrane to the esophageal tumor.

[0151] It is further noted that the patient's trachea and throat can tolerate heat being provided under positive pressure at an amount significantly MORE than temperature within a conventional sauna chamber. The LCD screen of the heat gun can provide the medical provider with ongoing temperatures of the air going into the trachea so that the patient is not harmed by excess heat than that programmed to treat the targeted esophageal tumor.

[0152] FIG. 9 is a diagrammatic side view in partial cross section of the throat, including the epiglottis, the larynx, vocal cords, trachea (windpipe), and esophagus (foodway), including the thin membranous trachea-esophageal party wall that separates the trachea from the esophagus, wherein, by heat conduction, inhaled heated air in the trachea passes through the membranous trachea-esophageal party wall by heat conduction to the esophagus which may have one or more esophageal cancerous tumors therein, and causes cell death of the esophageal cancer tumor, as well as to cause cell death of dysplasia pre-cancerous cells, such as known as Barrett's esophagus syndrome, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus.

[0153] It is further noted that preferably the health care practitioner, the patient and/or any assistant may wear protective safety eyeglasses.

[0154] In the foregoing description, certain terms and visual depictions are used to illustrate the preferred embodiment. However, no unnecessary limitations are to be construed by the terms used or illustrations depicted, beyond what is shown in the prior art, since the terms and illustrations are exemplary only, and are not meant to limit the scope of the present invention.

[0155] It is further known that other modifications may be made to the present invention, without departing the scope and spirit of the invention as noted in the appended Claims.

Claims

1. A method for treating esophageal cancer abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues in-situ in the esophagus of a patient with a flow of heated air that can be inhaled through a mask and mouthpiece, comprising the steps of: providing and directing the flow of the heated air from a heat source to be inspired/inhaled through the mouth and into the trachea/windpipe, at an elevated temperature less than the highest sauna air temperatures into the trachea of the patient at a sufficient constant positive pressure and temperature to uniformly heat the patient's tracheoesophageal party wall membrane extending between the esophagus and the respiratory trachea in the patient; monitoring and allowing the heated air to heat the patient's tracheoesophageal party wall membrane at a predetermined temperature and constant positive air pressure through the tracheoesophageal party wall membrane to conductively heat the esophageal cancer tumor or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, for a predetermined period of time adjacent to the heated trachea and the tracheoesophageal party wall membrane; whereby the heat provided at a predetermined temperature and at constant positive pressure from the heater air is inspired by the

patient through the trachea migrates/transmits/disperses conductively across the tracheoesophageal party wall membrane separating the patient trachea from the esophagus to contact the esophageal tumor or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, at the predetermined heat and positive pressure sufficient to cause cell death thereat.

2. The method of claim 1, wherein said heat source has an outlet nozzle which is sufficiently large and shaped to deliver heat through a mask to the esophageal cancer through the tracheoesophageal party wall membrane, separating the trachea from the esophagus.

3. The method of claim 1, wherein said heat source has a display comprising an LCD screen showing both of said temperatures, including said set programmed temperature.

4. The method of claim 3 in which the set programmed temperature is displayed on the screen of the heat source.

5. The method of claim 1, wherein where heat through heated air is administered to the esophageal tumor or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, located adjacent to the tracheoesophageal party wall membrane, separating the trachea from the esophagus at a range of about 107° F. to 180+° F.

6. The method of claim 5, wherein heat through heated air is administered to the esophageal cancer or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, located adjacent to the tracheoesophageal party wall membrane, separating the trachea from the esophagus at a range of about 107° F. to 130° F.

7. The method of claim 5, further comprising the step of inserting a subcutaneous needle probe temperature monitor into the in-situ site of the esophageal tumor or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, being treated and measuring the temperature of the tumor being treated.

8. The method of claim 1, where heat through heated air is administered by conduction across the tracheoesophageal party wall membrane to the esophageal tumor or abnormal esophageal pre-cancerous cells and Barrett's esophagus with dysplasia cells and tissues, as well as fungal, bacterial, protozoan, parasitic infections of any kind of the esophagus, for about one hour.

9. The method of claim 1, wherein the positive pressure applied to the heated air flow is provided at between about 3 psi to about 16 psi, or expressed in centimeters per unit of water, as being from 1-20 cm/H₂O.
